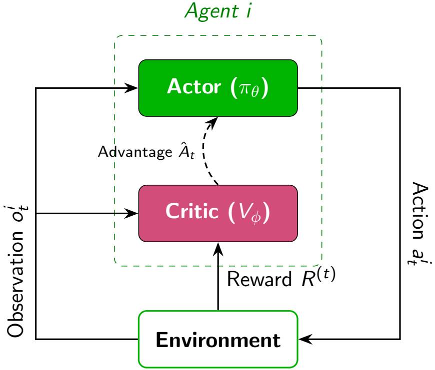
Theoretical Background: Multi-Agent Reinforcement Learning

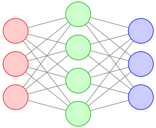
- **Decentralized Execution:** Agents independently decide actions based on local observations o^i_t.
- **Shared Environment:** Environment states (Draft, Global Memory) are updated by joint actions.
- **Cooperative Goal:** All agents maximize a shared global reward $R^{(t)}$.

Theoretical Background: Actor-Critic in MARS-RWG

Architecture



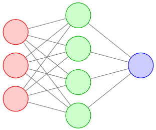
Actor Update (Gradient Ascent)



$$\theta \leftarrow \theta + \alpha \nabla_\theta \mathcal{L}^{\text{CLIP}}(\theta)$$

$$\mathcal{L}^{\text{CLIP}} = \mathbb{E}\left[\min\left(r_t \hat{A}_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right]$$

Critic Update (Gradient Descent)



$$\phi \leftarrow \phi - \beta \nabla_\phi \mathcal{L}^{\text{VF}}(\phi)$$

$$\mathcal{L}^{\text{VF}} = \mathbb{E}\left[\left(V_\phi(o_t^i) - V_t^{\text{target}}\right)^2\right]$$